

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1782

Harbor Beach 138 kV
February 17, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1782. The Interconnection Customer proposes a 205.7 MW natural gas combined cycle facility interconnecting at the Harbor Beach 138 kV substation in the Virginia area. The request was received on June 15, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 484 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,565,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Victory Way / Wheelabrator Portsmouth 138 kV	Wheelabrator Portsmouth / Gosport 138 kV L	122.4	234	11.8
Ne-Hobbs / Harbor Beach 138 kV L	Mira Vista / Yarnen 345 kV L	112.0	534	70.0
No Mans Land / Ne-Hobbs 138 kV	Mira Vista / Yarnen 345 kV	108.9	518	70.0
Thole St / Reeves Ave 138 kV	Mickens / Mira Vista 345 kV	104.9	224	5.0
Reeves Ave / Wheelabrator Portsmouth 138 kV line	Mira Vista / Mickens 345 kV Line	103.7	282	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wheelabrator Portsmouth / Victory Way Substation 138 kV	Rebuild line with higher-capacity conductor	\$9,552,000	17% (\$1,614,000)
Harbor Beach / N-Hobbs 138 kV L	Rebuild line with higher-capacity conductor	\$3,869,000	21% (\$800,000)
No Mans Land / Ne-Hobbs 138 kV	Reconductor line	\$980,000	21% (\$203,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Thole St / Reeves Ave 138 kV	Reconductor line	\$1,620,000	37% (\$595,000)
Reeves Ave / Wheelabrator Portsmouth 138 kV line	Reconductor line	\$960,000	37% (\$353,000)
	Total allocated to project		\$3,565,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.